

Heydar Soudani

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About Me

I am a final-year PhD candidate at Radboud University working on **trustworthy AI** for information-seeking tasks. My research spans **Retrieval-Augmented Generation**, **deep research agents**, and **uncertainty quantification**, with the goal of building systems that solve complex reasoning tasks and can signal when their answers should not be trusted. My work has appeared at ACL, EMNLP, SIGIR, ACM Computing Surveys, COLING, SIGIR-AP, and ICTIR.

Alongside my academic research, I collaborate with industry as an applied AI researcher; most recently on agentic search for the legal domain at Thomson Reuters, and on uncertainty quantification for production RAG systems at KPN.

Education

PhD	Computer Science , Radboud University, Nijmegen, Netherlands	Feb 2023 – Present
	• Title: Trustworthy Information-Seeking Systems • Supervisors: Faegheh Hasibi, Arjen P. de Vries	
MS	Computer Engineering , Sharif University, Tehran, Iran	Sep 2019 – Jun 2022
	• Title: Predicting Novelty Concepts in Data Stream • Supervisors: Hamid Beigy	
BS (Major)	Electrical and Electronics Engineering , Polytechnic of Tehran, Tehran, Iran	Sep 2014 – Sep 2019
	• Title: Providing new cloud services on the Kubernetes platform • Supervisors: Hassan Taheri	
BS (Minor)	Computer Engineering , Polytechnic of Tehran, Tehran, Iran	Sep 2016 – Sep 2018

Selected Publications

- [1] **Heydar Soudani**, Elizabeth Lingg, Faegheh Hasibi, Navid Rekabsaz. "When Deep Research Agents Stagnate: Enhancing Reasoning with Retrieval-Aware Agent Control" [🔗](#) *EMNLP 2026*.
- [2] **Heydar Soudani**, Hamed Zamani, Faegheh Hasibi. "Uncertainty Quantification for Retrieval-Augmented Reasoning" [🔗](#) *SIGIR 2026*.
- [3] Mahta Rafiee, **Heydar Soudani**, Zahra Abbasiantaeb, Mohammad Aliannejadi, Faegheh Hasibi, Hamed Zamani. "Total Recall QA: A Verifiable Evaluation Suite for Deep Research Agents" [🔗](#) *SIGIR 2026*.
- [4] Simon Binz, **Heydar Soudani**, Faegheh Hasibi. "Uncertainty Quantification for Multimodal Retrieval Augmented Generation" [🔗](#) *ICTIR 2026*.
- [5] **Heydar Soudani**, Roxana Petcu, Evangelos Kanoulas, Faegheh Hasibi. "A Survey on Recent Advances in Conversational Data Generation" [🔗](#) *ACM Computing Surveys 2026*.
- [6] **Heydar Soudani**, Evangelos Kanoulas, Faegheh Hasibi. "Why Uncertainty Estimation Methods Fall Short in RAG: An Axiomatic Analysis" [🔗](#) *ACL (Findings) 2025*.
- [7] **Heydar Soudani**, Evangelos Kanoulas, Faegheh Hasibi. "Fine Tuning vs. Retrieval Augmented Generation for Less Popular Knowledge" [🔗](#) *SIGIR-AP 2024*.
- [8] **Heydar Soudani**, Mohammad Hassan Mojab, Hamid Beigy. "Persian Natural Language Inference: A Meta-learning Approach" [🔗](#) *COLING 2022*.

Work Experience

Doctoral Researcher , Nijmegen, Netherlands	Radboud University Feb 2023 – Present
• Supervisors: Faegheh Hasibi, Arjen de Vries • Duties: Researching trustworthy generative information access, with a focus on	

deep research agents and uncertainty quantification. Published at ACL, SIGIR, ACM Computing Surveys, SIGIR-AP, and ICTIR.

Data Scientist Intern, Zug, Switzerland, Remote

- **Supervisors:** Navid Rekasaz, Elizabeth Lingg
- **Duties:** Researched and developed agentic search systems for the legal domain. Published at EMNLP.

Thomson Reuters
Nov 2025 – May 2026

Data Scientist Intern, Amsterdam, Netherlands, On-site

- **Supervisor:** Gianluigi Bardelloni
- **Duties:** Applied uncertainty quantification methods to internal RAG systems to make their responses more reliable.

KPN
May 2025 – Oct 2025

Web Developer, Andishe Fartak Amirkabir (Atrovan), Part-time

- **Duties:** Developed web applications for diverse domains, including a smart metering system (AtroMeter), a fleet management system (Navgoon), a home automation platform (Smart Home), and a building management system (BMS)

Atrovan, Tehran, Iran
Dec 2017 – Jun 2022

Research Assistant, Control of Multi Vehicle Systems Laboratory (CMVL)

- **Supervisor:** Farzaneh Abdollahi
- **Duties:** Collaborated on AVR and Raspberry Pi programming, computer vision for localization, and PCB design with Altium Designer.

Polytechnic of Tehran
Jan 2016 – Oct 2017

Tutorials, Talks & Organizing

Workshop, DIR 2025. [The 22nd Dutch-Belgian Information Retrieval Workshop](#)

Nijmegen, Netherlands
Oct 2025

Doctoral consortium, SIGIR 2025. [Enhancing Knowledge Injection in Large Language Models for Efficient and Trustworthy Responses](#)

Padua, Italy
July 2025

Tutorial, WebConf 2024. [Data Augmentation for Conversational AI \(second edition\)](#)

Singapore, Singapore
May 2024

Tutorial, CIKM 2023. [Data Augmentation for Conversational AI \(first edition\)](#)

Birmingham, UK
Oct 2023

Languages

Persian: Native **Arabic:** Native **English:** Fluent

Programming Skills

Languages: Python JavaScripts C++ C

Libraries/Frameworks: PyTorch TensorFlow HuggingFace NumPy Pandas Matplotlib

Tools: $\text{\LaTeX}$ Git

References

Faegheh Hasibi, Assistant Professor, Radboud University, f.hasibi@cs.ru.nl

Evangelos Kanoulas, Professor, University of Amsterdam, e.kanoulas@uva.nl